

MATHEMATICAL INDUCTION



SYNOPSIS

➤ Principle of Finite Mathematical

Induction :

For $n \in \mathbb{N}$, let $P(n)$ be a statement in terms of n .
If $P(1)$ is true and $P(k)$ is true $\Rightarrow P(k+1)$ is true,
then $P(n)$ is true, for all $n \in \mathbb{N}$.

➤ Principle of Complete Mathematical

Induction :

For $n \in \mathbb{N}$, let $P(n)$ be a statement in terms of n .
If $P(1), P(2), P(3), \dots, P(k-1)$ are true $\Rightarrow P(k)$ is true,
then $P(n)$ is true, for all $n \in \mathbb{N}$.

➤ $a, (a+d), (a+2d), \dots$ form an A.P. then

(i) n^{th} term $t_n = a + (n-1)d$, Where a is the first term and d is the common difference.

(ii) Sum of n terms $S_n = \frac{n}{2}[2a + (n-1)d]$

$$= \frac{n}{2}[a + l]$$

Where a = first term, l = last term

➤ $a, ar, ar^2, \dots$ form a G.P then

(i) n^{th} term $t_n = ar^{n-1}$, Where a = first term
 r = common ratio

(ii) Sum of n terms $S_n = a \frac{(r^n - 1)}{r - 1}$;

(iii) In an infinite G.P, Sum of Infinite terms is

$$S_\infty = \frac{a}{1-r} \text{ where } |r| < 1$$

➤ $a + (a+d)r + (a+2d)r^2 + \dots$

$\dots + [a + (n-1)d]r^{n-1}$ form A.G.P. then

(i) n^{th} term $t_n = [a + (n-1)d]r^{n-1}$

(ii) Sum of n terms

$$S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{(1-r)}$$

(iii) Sum of Infinite terms

$$S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2} \text{ where } |r| < 1$$

W.E-1: Sum to infinite terms of

$$1 + 3x + 5x^2 + 7x^3 + \dots + \infty \text{ (If } |x| < 1 \text{) is } \underline{\hspace{2cm}} \quad [\text{Eam-1998}]$$

Sol: $s_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

where $a = 1, d = 2, r = x$

$$= \frac{1}{1-x} + \frac{2x}{(1-x)^2} = \frac{1+x}{(1-x)^2}$$

Similarly If $|x| < 1$,

$$s_\infty = 1 + 2x + 3x^2 + 4x^3 + \dots + \infty = \frac{1}{(1-x)^2}$$

Some important points:

i) Sum of first n natural numbers i.e.

$$\sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}, \forall n \in \mathbb{N}$$

ii) Sum of the squares of first n natural numbers is

$$\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}, \forall n \in \mathbb{N}$$

iii) Sum of the cubes of first n natural numbers is

$$\sum n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$= \left(\sum n\right)^2, \forall n \in \mathbb{N}$$

iv) $1^4 + 2^4 + 3^4 + \dots + n^4 = \sum n^4$

$$= \frac{n(n^4 - 1)}{5} + \frac{n(n+1)(3n^2 - n + 1)}{6}$$

v) Sum of the first 'n' odd +ve integers =

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

vi) Sum of the first 'n' even +ve integers =

$$2 + 4 + 6 + \dots + 2n = n(n+1)$$

vii) $\sum n^2 \neq (\sum n)^2$... etc.,

$$\text{viii) } \sum n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$$\text{ix) } \sum n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

x) $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots$ 'n' terms

$$= -\frac{n(n+1)}{2}; \text{ if n is even}$$

$$= \frac{n(n+1)}{2}; \text{ if n is odd}$$

$$\text{xi) } (n+3)^2 \leq 2^{n+3}$$

xii) The sum in the n^{th} bracket of

$$1 + (2+3) + (4+5+6) + \dots \text{ is } \frac{n(n^2+1)}{2}$$

$$\text{xiii) } \sum_{k=1}^n k \left(1 + \frac{1}{n}\right)^{k-1} = n^2$$

➤ The inequality

i) $2^n < n!$ is true for all $n \geq 4$

ii) $2^n > 2n+1$ is true for all $n \geq 3$

iii) $2^n \leq (n+1)!$ is true for all $n \in \mathbb{N}$

iv) $2n-3 \leq 2^{n-2}$ is true for all $n \geq 5$

W.E-2: Sum to n terms of $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$ is

Sol:
$$s_n = \frac{1}{3} \left[1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \dots + \frac{1}{3n-2} - \frac{1}{3n+1} \right]$$

$$= \frac{1}{3} \left[1 - \frac{1}{3n+1} \right] = \frac{n}{3n+1}$$

➤ $\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + (n-1)\beta)$

$$= \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \times \sin\left[\alpha + (n-1)\frac{\beta}{2}\right]$$

➤ $\cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \dots + \cos(\alpha + (n-1)\beta)$

$$= \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \times \cos\left[\alpha + (n-1)\frac{\beta}{2}\right]$$

➤ $\cos \alpha \cdot \cos 2\alpha \cdot \cos 4\alpha \dots \cos(2^{n-1}\alpha) = \frac{\sin(2^n \alpha)}{2^n \sin \alpha}$

➤ $\left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right) \dots \left(1 + \frac{2n+1}{n^2}\right) = (n+1)^2$

➤ For a sequence $T_1, T_2, T_3, T_4, \dots$, the difference of two consecutive terms $T_2 - T_1, T_3 - T_2, T_4 - T_3, \dots$ are in A.P or G.P then n^{th} term of given series are in the form of $an^2 + bn + c$ or $ar^n + b$ where a, b, c to be determined

Some important points:

i) The sum of cubes of three consecutive natural numbers is always divisible by 9

ii) For all positive integral values of n, $x^n - y^n$ is divisible by $x - y$.

iii) For all positive integral values of n, $x^{2n+1} + y^{2n+1}$ is divisible by $x + y$.

iv) For all positive integral values of n, $x^{2n-1} + y^{2n-1}$ is divisible by $x + y$.

v) $P^{n+1} + (P+1)^{2n-1}$ is divisible by

$$P^2 + P + 1, \forall n \geq 2$$

vi) $n^p - n$ is divisible by $P \forall n \geq 2$ where P is prime.

vii) n is any odd integer then $n(n^2 - 1)$ is divisible by 24.

viii) The product of " n " consecutive natural numbers is always divisible by $n!$.

- If x, y, m are positive integers then x is said to be congruent of y modulo m if $x - y$ is divisible by m and is denoted by $x \equiv y \pmod{m}$



C.U.Q

1. Mathematical Induction is the principle containing the set

1) R 2) N 3) Q 4) Z

2. Let $P(n)$ be a statement and let

$P(n) \Rightarrow P(n+1) \forall n \in N$, then $P(n)$ is true.

- 1) for all ' n ' 2) for all $n > 1$
3. for all $n > m$, m being a fixed +ve integer
4. Nothing can be said

3. The statement $P(n)$

$(1 \times 1!) + (2 \times 2!) + (3 \times 3!) + \dots$

$\dots + (n \times n!) = (n+1)! - 1$ is

- 1) True for all $n > 1$ 2) Not true for any n
3) True for all $n \in N$ 4) True for all $n > 10$.

4. The smallest +ve integer n for which

$n! < \left(\frac{n+1}{2}\right)^n$ holds is

- 1) 1 2) 2 3) 3 4) 4

5. Let $S(K) = 1 + 3 + 5 + \dots + (2K - 1) = K^2 + 3$. Then which of the following is true?

- 1) $S(1)$ is correct 2) $S(K) \Rightarrow S(K+1)$
3) $S(K) \neq S(K+1)$
4) P.O.F.M.I can be used to prove the formula

6. For every natural number n , $n(n+1)$ is always

- 1) even 2) odd
3) multiple of 3 4) multiple of 4

7. Let $P(n)$ denote the statement that $n^2 + n$ is odd. It is seen that $P(n) \Rightarrow P(n+1)$, $P(n)$ is true for all

1) $n > 1$ 2) n 3) $n > 2$ 4) no value of n

8. $\{n(n+1)(2n+1) : n \in Z\} \subset$

- 1) $\{6k : k \in Z\}$ 2) $\{12k : k \in Z\}$
3) $\{18k : k \in Z\}$ 4) $\{24k : k \in Z\}$

9. If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + \dots}}}$ having ' n ' radical signs then by methods of mathematical induction which is true

- 1) $a_n > 7, \forall n \geq 1$ 2) $a_n < 2, \forall n \geq 1$
3) $a_n < 7, \forall n \geq 1$ 4) $a_n > 13, \forall n \geq 1$

10. If $a > 0, x > 0$, then

$$\frac{1}{\sqrt{a} + \sqrt{a+x}} + \frac{1}{\sqrt{a+x} + \sqrt{a+2x}} + \dots + \frac{1}{\sqrt{a+(n-1)x} + \sqrt{a+nx}} =$$

- 1) $\frac{\sqrt{a+nx} + \sqrt{a}}{x}$ 2) $\frac{\sqrt{a+nx} - \sqrt{a}}{x}$
3) $\frac{x}{\sqrt{a+nx} + \sqrt{a}}$ 4) $\frac{x}{\sqrt{a+nx} - \sqrt{a}}$

11. $\forall n \in N, \frac{n^4}{24} + \frac{n^3}{4} + \frac{11n^2}{24} + \frac{n}{4}$ is a

- 1) Rational Number 2) Integer
3) Natural Number 4) Real Number

12. Let $P(n) : a^n + b^n$ such that a, b are even, then $P(n)$ will be divisible by $a+b$ if

- 1) $n > 1$ 2) $\forall n$ is odd
3) $\forall n$ is even 4) $n > 2$

13. If p is a prime number, then $n^p - n$ is divisible by p when n is a

- 1) Natural number greater than 1
2) Irrational number
3) Complex number 4) Odd number

14. The greatest +ve integer which divides

$(n+1)(n+2)\dots(n+r)$, for all $n \in N$ is

- 1) $(r+1)!$ 2) $r!$ 3) r 4) $r-1$

C.U.Q-KEY

- | | | | |
|-------|-------|-------|-------|
| 01) 2 | 02) 4 | 03) 3 | 04) 2 |
| 05) 2 | 06) 1 | 07) 4 | 08) 1 |
| 09) 3 | 10) 2 | 11) 3 | 12) 2 |
| 13) 1 | 14) 2 | | |

C.U.Q-HINTS

- By the definition of P.O.F.M.I
- By the definition of P.O.F.M.I
- When $n=1$, $L.H.S.=1.1!$, $R.H.S.=2!-1!$
when $n=2$, $L.H.S.=1.1!+2.2!$, $R.H.S.=3!-1!$
Hence $P(n)$ is true $\forall n \in N$
- By Verification
- By verification
- The product of 2 consecutive numbers is always even.
- It is obvious.
- Put $n=1, 2$ and verify the options.
- $a_1 = \sqrt{7} < 7$; Let $a_m < 7$. Then $a_{m+1} < \sqrt{7+a_m}$
 $a_{m+1}^2 = 7 + a_m < 7 + 7 = 14$
 $a_{m+1} < \sqrt{14} < 7$; So $a_n < 7, \forall n$
- By verification
- Put $n=1, n=2$ and verify the options.
- $P(n) = a^n + b^n \forall n \in N$
put $n=1$, $\therefore P(1) = a+b$ which is divisible by $a+b$
put $n=2$, $\therefore P(2) = a^2 + b^2$ not divisible by $a+b$, put $n=3$
 $\therefore P(3) = a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
which is divisible by $a+b$.
With the help of induction we conclude that $P(n)$ will be divisible by $a+b$ if n is odd.

13. put $n=4$ and $P=2$.

14. Product of r successive integers is divisible by $r!$



LEVEL - I (C.W)

Principle of Mathematical Induction:

- A student was asked to prove a statement by induction. He proved (i) $P(5)$ is true and (ii) truth of $P(n) \Rightarrow$ truth of $P(n+1)$, $n \in N$. On the basis of this, he could conclude that $P(n)$ is true
1) for no $n \in N$ 2) for all $n \geq 5, n \in N$
3) for all $n \in N$ 4) for all $n \geq 1, n \in N$
- If $P(n)$ be the statement $n(n+1)+1$ is an integer, then which of the following is even
1) $P(2)$ 2) $P(3)$
3) $P(4)$ 4) None of the above
- $n > 1$, n even $\Rightarrow$ digit in the units place of $2^{2n} + 1$
1) 5 2) 7 3) 6 4) 1
- $\log(x)^n = n \cdot \log x$ is true for n .
1) $\forall n \in N$ 2) $\forall n \in Z$
3) n is positive odd integer
4) n is positive even integer
- Inequalities:**
 $(n!)^2 > n^n$ is true for
1) $\forall n \in N$ 2) $\forall n > 1, n \in N$
3) $\forall n > 2, n \in N$ 4) $\forall n \in Z$
- Let $P(n): 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$ is true for
1) $\forall n \in N$ 2) $n=1$
3) $n > 1, \forall n \in N$ 4) $n > 2$
- Summation of Series:**
 $1^3 - 2^3 + 3^3 - 4^3 + \dots + 9^3 =$
1) 425 2) -425 3) 475 4) -475
- $\frac{1}{2.5} + \frac{1}{5.8} + \frac{1}{8.11} + \dots n \text{ terms} =$

- 1) $\frac{n}{6n+4}$ 2) $\frac{n}{3n+2}$
 3) $\frac{n}{4n+6}$ 4) $\frac{1}{2(2n+3)}$
 9. $2+7+14+\dots+(n^2+2n-1) =$
 1) $\frac{n(2n^2+9n+1)}{6}$ 2) $\frac{2n^2+9n+1}{6}$
 3) $\frac{2n^2+9n+1}{12}$ 4) $\frac{2n^2+9n+1}{24}$
 10. $1+3+6+10+\dots+\frac{(n-1)n}{2}+\frac{n(n+1)}{2} =$
 1) $\frac{n(n+1)(n+2)}{3}$ 2) $\frac{(n+1)(n+2)}{6}$
 3) $\frac{n(n+1)(n+2)}{6}$ 4) $\frac{(n+2)(n+1)^2}{3}$
 11. $3.6+6.9+9.12+\dots+3n(3n+3) =$
 1) $\frac{n(n+1)(n+2)}{3}$ 2) $3n(n+1)(n+2)$
 3) $\frac{(n+1)(n+2)(n+3)}{3}$ 4) $\frac{(n+1)(n+2)(n+4)}{4}$
 12. $1.6+2.9+3.12+\dots+n(3n+3) =$
 1) $n(n+1)(n+2)$ 2) $(n+1)(n+2)(n+3)$
 3) $(n+2)(n+3)(n+4)$ 4) $(n-1)n(n+1)$
 13. $1^3+1^2+1+2^3+2^2+2+3^3+3^2+3+\dots 3n$ terms =
 1) $\frac{n(n+1)(n^2+12n+5)}{12}$
 2) $\frac{n(n+1)(3n^2+7n+8)}{12}$
 3) $\frac{n(n+1)(n+2)(n^2+5n+6)}{12}$
 4) $\frac{(n+1)(n+2)(n+3)}{4}$
 14. $\frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+\dots(n-3)$ terms
 1) $\frac{n}{n+2}$ 2) $\frac{n+1}{n(n+5)}$
 3) $\frac{n-3}{2n-5}$ 4) $\frac{n-1}{n(2n-3)}$
 15. $1+3+7+15\dots n$ terms =
 1) $2^{n+1}-n-2$ 2) n^2+n-2
 3) 2^n+n^2-2 4) n^2+n
 16. $2.4+4.7+6.10+\dots(n-1)$ terms =

- 1) $2n^3-2n^2$ 2) $\frac{n^3+3n^2+1}{6}$
 3) $2n^3+2n$ 4) $2n^3-n^2$
 17. $1^2+3^2+5^2+\dots$ upto n terms =
 1) $\frac{n(2n-1)(2n+1)}{6}$ 2) $\frac{n(2n-1)(2n+1)}{3}$
 3) $\frac{n(n+1)(2n+1)}{12}$ 4) $\frac{n(n+1)(2n-1)}{6}$
 18. Sum to n terms of the series
 $1+(1+x)+(1+x+x^2)+\dots$
 1) $\frac{n}{1-x}-\frac{x(1-x^n)}{(1-x)^2}$ 2) $\frac{n}{1-x}+\frac{x(1-x^n)}{(1-x)^2}$
 3) $\frac{n}{1-x}+\frac{x(1+x^n)}{(1-x)^2}$ 4) $\frac{-n}{1-x}+\frac{x(1-x^n)}{(1-x)^2}$
 19. If a, b and n are natural numbers then
 $a^{2n-1}+b^{2n-1}$ is divisible by
 1) $a-b$ 2) a^3+b^3 3) $a+b$ 4) a^2+b^2
 20. The number $a_n = 6^n - 5n$ for
 $n = 1, 2, 3, \dots$ when divided by 25
 leave the remainder
 1) 9 2) 7 3) 3 4) 1
Divisibility:
 21. If 'a' and 'b' are natural numbers such that
 $a^2 - b^2$ is prime number then $a^2 - b^2$ equals
 1) $a+b$ 2) $a-b$ 3) ab 4) 1
 22. $\forall n \in N, n^2(n^4-1)$ is divisible by
 1) 60 2) 120 3) 45 4) 90
 23. $\forall n \in N, 49^n + 16n - 1$ is divisible by
 1) 64 2) 49 3) 132 4) 32
 24. $\forall n \in N, 7^{2n} + 3^{n-1} \cdot 2^{3n-3}$ is divisible by
 1) 50 2) 25 3) 2425 4) 2550
 25. For all integers $n \geq 1$, which of the following
 is divisible by 9?
 1) $8^n + 1$ 2) $4^n - 3n - 1$
 3) $3^{2n} + 3n + 1$ 4) $10^n + 1$

LEVEL-I (C.W)-KEY

01) 2	02) 4	03) 2	04) 3
05) 3	06) 3	07) 1	08) 1
09) 1	10) 3	11) 2	12) 1
13) 2	14) 3	15) 1	16) 1
17) 2	18) 1	19) 3	20) 4
21) 1	22) 1	23) 1	24) 2
25) 2			

LEVEL-I (C.W)-HINTS

- By Verification
- $P(n) = n(n+1)+1$, $P(2) = 6+1 = 7$
 $P(3) = 3 \times 4 + 1 = 13$, $P(4) = 4 \times 5 + 1 = 21$
 None of the above is even
- Put $n = 2$, $2^4 + 1 = 17$.
- By Verification
- By Verification
- For $n = 1$, L.H.S. of $P(1) = 1$
 R.H.S. of $P(1) = 2 - 1 = 1$, L.H.S. = R.H.S.
 $\therefore 1$ can not be < 1 $\therefore P(1)$ is not true.
 as L.H.S. of $P(1) = 1 < 1$ (false)
 again $n = 2$
 L.H.S. of $P(2) = 1 + 1/4 = 5/4$
 R.H.S. of $P(2) = 2 - 1/2 = 3/2 = 6/4$
 $\therefore$ L.H.S. $P(2) < \text{R.H.S. of } P(2)$
 $\therefore P(2)$ is true
- $\sum_{n=1}^9 n^3 - 2(2^3 + 4^3 + 6^3 + 8^3)$
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n - 3 = 1$, $S = \frac{1}{1.3} = \frac{1}{3}$
- Put $n = 2$ and verify the options.

- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options
- $\frac{1-x}{1-x} + \frac{1-x^2}{1-x} + \frac{1-x^3}{1-x} + \dots + \frac{1-x^n}{1-x}$
- Put $n = 2$ and verify the options
- Put $n = 2$ and verify the options
- We know that $a^2 - b^2 = (a+b)(a-b)$ given
 $a^2 - b^2$ is a prime $\therefore a^2 - b^2$ is either divisible by 1 or the number itself
 $\therefore a^2 - b^2 = a + b$ ($\because a + b \neq 1$ & $a - b = 1$
 as $a, b \in N$)
- Put $n = 2$ and verify the options.
- Put $n = 1$ and verify the options.
- Put $n = 1, 2$ and verify the options.
- Using substitution verify the options

LEVEL - I (H.W)

Summation of Series:

- $1+4+10+19+\dots+\frac{3n^2-3n+2}{2}$
 1) $\frac{n^2(n^2+1)}{2}$ 2) $\frac{n(n^2+1)}{2}$
 3) $\frac{n^2(n+1)}{2}$ 4) $\left\{\frac{n(n+1)}{2}\right\}^2$
- $1.3+3.5+5.7+\dots+(2n-1)(2n+1)=$
 1) $\frac{n(4n^2+6n-1)}{3}$ 2) $\frac{n(3n^2+5n+1)}{3}$
 3) $\frac{n(5n^2+7n-1)}{3}$ 4) $\frac{n(7n^2-5n+1)}{3}$
- $1.4+2.5+\dots+n(n+3)=$
 1) $\frac{n(n+3)(n+5)}{9}$ 2) $\frac{n(n+1)(n+5)}{3}$
 3) $\frac{n(n+5)(n+7)}{6}$ 4) $\frac{n(n+3)(n+9)}{12}$
- $2+3+5+6+8+9+\dots+2n$ terms =
 1) $3n^2+2n$ 2) $4n^2+2n$
 3) $4n^2$ 4) $5n^2+2n$
- n^{th} term of the series $4+14+30+52+\dots=$

- 1) $5n - 1$ 2) $2n^2 + 2n$
 3) $3n^2 + n$ 4) $2n^2 + 2$
6. $2.3+3.4+4.5+\dots+n$ terms =
 1) $\frac{n(n^2+6n+11)}{6}$ 2) $\frac{n(n^2+6n+14)}{9}$
 3) $\frac{n(n^2+6n+11)}{3}$ 4) $\frac{n(n^2+6n+17)}{12}$
7. The n th term of the series $3+7+13+21+\dots$ is
 1) $4n - 1$ 2) $n^2 + 2n$ 3) $n^2 + n + 1$ 4) $n^2 + 2$
8. The value of
 $1.2.3+2.3.4+3.4.5+\dots+n$ terms is
 1) $\frac{n(n+1)(n+2)(n+3)}{12}$
 2) $\frac{n(n+1)(n+2)(n+3)}{3}$
 3) $\frac{n(n+1)(n+2)(n+3)}{4}$
 4) $\frac{(n+2)(n+3)(n+4)}{6}$
9. $3.6+4.7+5.8+\dots+(n-2)$ terms =
 1) $n^3 + n^2 + n + 2$
 2) $\frac{2n^3+12n^2+10n-84}{6}$
 3) $2n^3+12n^2+10n+84$
 4) $2n^3-12n^2+10n-84$
10. $2+3.2+4.2^2+\dots$ upto n terms =
 1) $(n+1)2^{n-1}$ 2) $n2^{n+1}$
 3) $n2^n$ 4) $(n+1)2^n$
11. If $1+5+12+22+35+\dots$ to n terms = $\frac{n^2(n+1)}{2}$ then n^{th} term of L.H.S is
 1) $\frac{n(4n-1)}{3}$ 2) $\frac{n(3n-1)}{2}$

- 3) $\frac{n(3n+1)}{2}$ 4) $\frac{n(4n+1)}{3}$
12. $2.4+4.7+6.10+\dots$ upto $(n-1)$ terms =
 1) $2n^3+2n^2$ 2) $\frac{1}{6}(n^3+3n^2+1)$
 3) $2n^3+2n$ 4) $2n^3-2n^2$
13. $\frac{1}{4.7}+\frac{1}{7.10}+\frac{1}{10.13}+\dots$ upto n terms =
 1) $\frac{n}{4(4n+3)}$ 2) $\frac{n}{4(6n+1)}$
 3) $\frac{n}{4(3n+4)}$ 4) $\frac{n}{4(3n-4)}$
- Divisibility:**
14. Sum of the cubes of three successive natural number is divisible by
 1) 9 2) 27 3) 54 4) 99
15. $\forall n \in \mathbb{N}, 3^{2n} + 7$ is divisible by
 1) 8 2) 16 3) 24 4) 64
16. $\forall n \in \mathbb{N}, 7.5^{2n} + 12.6^n$ is divisible by
 1) 13 2) 19 3) 247 4) 26
17. For every natural number n , $3^{2n+2} - 8n - 9$ is divisible by
 1) 16 2) 128 3) 256 4) 512
18. For every integer $n \geq 1$, $(3^{2^n} - 1)$ is divisible by
 1) 2^{n+2} 2) 2^{n+1} 3) 2^{n+3} 4) 2^n
19. Use mathematical induction, to find $41^n - 14^n$ is a multiple of
 1) 2, 8 2) 27 3) 26 4) 14

LEVEL - I (H.W)-KEY

- | | | | |
|-------|-------|-------|-------|
| 01) 2 | 02) 1 | 03) 2 | 04) 1 |
| 05) 3 | 06) 3 | 07) 3 | 08) 3 |
| 09) 2 | 10) 3 | 11) 2 | 12) 4 |
| 13) 3 | 14) 1 | 15) 1 | 16) 2 |
| 17) 1 | 18) 1 | 19) 2 | |

LEVEL - I (H.W)-HINTS

- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 1$ and verify the options.
L.H.S = $2 + 3$
R.H.S = $3(1)^2 + 2(1) = 5$
- Given $T_2 = 14$
Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 3$ and verify the options.
- By verification
- Put $n = 3$ and verify the options.
- Put $n = 2$ and verify the options
- Put $n = 2$ verify the options
- Put $n - 1 = 2$ and verify the options
- $\frac{1}{3} \left[\frac{1}{4} - \frac{1}{7} + \frac{1}{7} - \frac{1}{10} + \dots + \frac{1}{3n+1} - \frac{1}{3n+4} \right]$
 $= \frac{n}{4(3n+4)}$
- $1^3 + 2^3 + 3^3 = 36$ divisible by 9.
- Put $n = 1$, $3^2 + 7 = 16$
Put $n = 2$, $3^4 + 7 = 88$, G.C.D. of 16 & 88 = 8
- Put $n = 1$, $7.5^2 + 12.6 = 247$
Put $n = 2$, $7.5^4 + 12.6^2 = 4807$
G.C.D. of 247 & 4807 = 19
- $3^{2n+2} - 8n - 9, \forall n \in \mathbb{N}$
put $n = 2$, $f(2) = 704$, divisible by 16
- Let $P(n) = 3^{2^n} - 1$
 $P(1) = 3^{2^1} - 1 = 8 = 1.8$ is divisible by 2^3 .
 $P(2) = 3^{2^2} - 1 = 80 = 5.2^4$ is divisible by 2^4 .
- Verification



LEVEL - II (C.W)

Finite Mathematical Induction,
Summation of Series:

- If $2^3 + 4^3 + 6^3 + \dots + (2n)^3 = Kn^2(n+1)^2$
then $k =$

- 1) $1/2$ 2) 1 3) $3/2$ 4) 2

- If $a_k = \frac{1}{k(k+1)}$ for $k = 1, 2, 3, \dots, n$, then

$$\left(\sum_{k=1}^n a_k \right)^2 =$$

- 1) $\frac{n}{n+1}$ 2) $\frac{n^2}{(n+1)^2}$ 3) $\frac{n^4}{(n+1)^4}$ 4) $\frac{n^6}{(n+1)^6}$

- $\sum_{n=0}^{\infty} (-1)^n x^{n+1} =$

- 1) $\frac{x^n}{2(1+x)}$ 2) $\frac{x}{1+x}$ 3) $\frac{x}{x-1}$ 4) $\frac{x^n}{x-1}$

- $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos \{(n-1)\theta\}$
 $+ \cos n\theta =$

- 1) $\frac{\cos \left\{ \frac{1}{2}(n+1)\theta \right\} \cdot \sin \left(\frac{n\theta}{2} \right)}{\sin \left(\frac{\theta}{2} \right)}$ 2) $\frac{\cos(n+1)\theta}{\sin \left(\frac{\theta}{2} \right)}$

- 3) $\frac{\cos \left(\frac{(n-1)\theta}{2} \right) \sin \left(\frac{n\theta}{2} \right)}{\sin \frac{\theta}{2}}$

- 4) $\frac{\sin \left(\frac{n\theta}{2} \right) \cdot \cos \{(n+1)\theta\}}{\sin \left(\frac{\theta}{2} \right)}$

- If 'n' is a positive integer, then

$$n.1 + (n-1).2 + (n-2).3 + \dots + 1.n =$$

- 1) $\frac{n(n+1)}{2}$ 2) $\frac{n(n+1)(n+2)}{6}$

- 3) $\frac{(n+1)(n+2)}{2}$ 4) $\frac{n(n+1)(2n+1)}{6}$

- $1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots + n$
brackets =

- 1) $\frac{n(n+1)^2(n+2)^2}{12}$ 2) $\frac{n(n+1)^2(n+2)}{12}$

$$3) \frac{n^2(n+1)(n+2)}{12} \quad 4) \frac{(n+1)}{2}$$

$$7. \frac{1}{1^3} \cdot \frac{2}{2} + \frac{2}{1^3+2^3} \cdot \frac{3}{2} + \frac{3}{1^3+2^3+3^3} \cdot \frac{4}{2} + \dots \text{ n terms}$$

$$1) \frac{n^2}{(n+1)^2} \quad 2) \frac{n^3}{(n+1)^3} \quad 3) \frac{n}{n+1} \quad 4) \frac{1}{n+1}$$

$$8. \text{ If } t_n = \sum_1^n n, \text{ then } t_n^1 = \sum_1^n t_n =$$

$$1) \frac{n(n+1)}{2} \quad 2) \frac{n(n+3)}{2}$$

$$3) \frac{n(n+1)(n+2)}{6} \quad 4) \frac{n(n+4)}{3}$$

9. Let the statement $m^2 > 100$, the statement $P(k+1)$ will be true if

$$1) P(1) \text{ is true} \quad 2) P(2) \text{ is true}$$

$$3) P(k) \text{ is true} \quad 4) \text{ none of these}$$

10. For any integer $n \geq 1$, then

$$\sum_{k=1}^n k(k+2) \text{ is equal to} \quad [\text{Eam-2008}]$$

$$1) \frac{n(n+1)(n+2)}{6} \quad 2) \frac{n(n+1)(2n+1)}{6}$$

$$3) \frac{n(n+1)(2n+7)}{6} \quad 4) \frac{n(n+1)(2n+9)}{6}$$

11. $1 + \frac{1}{2}(1+2) + \frac{1}{3}(1+2+3) + \frac{1}{4}(1+2+3) + \dots$
upto 20 terms is

$$1) 110 \quad 2) 111 \quad 3) 115 \quad 4) 116$$

12. If $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \dots \text{ n terms} = \frac{kn}{n+1}$
then $k =$ [Eam-2012]

$$1) \frac{1}{4} \quad 2) \frac{1}{2} \quad 3) 1 \quad 4) \frac{1}{8}$$

Divisibility:

13. The greatest positive integer which divides $n(n+1)(n+2) \dots (n+r-1)$, $\forall n \in N$ is

$$1) r! \quad 2) (r+1)! \quad 3) n+r \quad 4) n-r+1$$

14. If $10^n + 3 \cdot 4^n + x$ is divisible by 9 for all $n \in N$, then least positive value of 'x' is

$$1) 1 \quad 2) 5 \quad 3) 14 \quad 4) 23$$

Inequalities:

15. If $n > 1$ then

$$1) \frac{(2n)!}{(n!)^2} = \frac{4n}{2n+1} \quad 2) \frac{(2n)!}{(n!)^2} < \frac{4n}{2n+1}$$

$$3) \frac{(2n)!}{(n!)^2} > \frac{4n}{2n+1} \quad 4) \text{ none}$$

16. $\forall n \in N$,

$$1) |\sin(nx)| < |\sin x| \quad 2) |\sin(nx)| < n |\sin x|$$

$$3) |\sin(nx)| \leq n |\sin x| \quad 4) \sin(nx) \leq \sin n$$

LEVEL - II (C.W)-KEY

01) 4	02) 2	03) 2	04) 1
05) 2	06) 2	07) 3	08) 3
09) 3	10) 3	11) 3	12) 1
13) 1	14) 2	15) 3	16) 3

LEVEL - II (C.W)-HINTS

$$1. \quad 2^3 (1^3 + 2^3 + 3^3 + \dots + n^3) = kn^2 (n+1)^2$$

2. Put $n = 1$ and verify the options.

$$3. \quad x(1-x+x^2-x^3+\dots) = x(1+x)^{-1}$$

4. Put $n = 1$ and verify the options.

5. Put $n = 2$ and verify the options.

6. Put $n = 2$ and verify the options.

7. Put $n = 2$ and verify the options.

$$8. \quad \sum_1^n t_n = \frac{1}{2} [\sum n^2 + \sum n]$$

9. $P(r)$ is true

$$\Rightarrow r^2 > 100 \Rightarrow r^2 + 2r + 1 > 100 + 2r + 1$$

$$\Rightarrow (r+1)^2 > 100 \Rightarrow P(k+1) \text{ is true.}$$

$P(k+1)$ is true when every $P(k)$ is so.

10. Put $n = 2$ and verify the options

$$11. t_n = \frac{1}{n}(1+2+3+\dots+n) = \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$\text{Now find } \sum_{i=1}^n \left(\frac{n+1}{2} \right) = \frac{1}{2} \frac{n(n+1)}{2} + \frac{n}{2} = S_n$$

now put $n = 20$

$$12. \frac{kn}{n+1} = \left[\frac{1}{2.4} + \frac{1}{4.6} + \frac{1}{6.8} + \dots + n \text{ terms} \right]$$

$$= \frac{1}{2} \left[\frac{4-2}{2.4} + \frac{6-4}{4.6} + \frac{8-6}{6.8} + \dots + \frac{1}{2n(2n+2)} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} + \frac{1}{4} - \frac{1}{6} + \frac{1}{6} - \frac{1}{8} + \dots + \frac{1}{2n} - \frac{1}{2n+2} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{2(n+1)} \right] = \frac{n}{4(n+1)} \Rightarrow k = \frac{1}{4}$$

13. Let $P(n) = n(n+1)(n+2)\dots(n+r-1)$
 $= (n+0)(n+1)(n+2)\dots(n+(r-1))$
 $= \text{Product of } r \text{ consecutive integers}$

$$14. n = 1 \Rightarrow 10 + 3.4 + x = 9m \Rightarrow x = 5$$

$$15. \text{ If } n = 2 \text{ then } \frac{(2n)!}{(n!)^2} = \frac{24}{4} = 6, \frac{4n}{2n+1} = \frac{8}{3}$$

$$\Rightarrow \frac{(2n)!}{(n!)^2} > \frac{4n}{2n+1}$$

$$16. |\sin nx| \leq 1 \leq n |\sin x|$$



LEVEL - II (H.W)

$$1. 4^3 + 5^3 + 6^3 + \dots + 10^3 =$$

1) 1905 2) 2358 3) 2447 4) 2989

$$2. \left(\sum n^3 \right) \left(\sum n \right) = \left(\sum n^2 \right)^2 \text{ if}$$

1) $n = 3$ 2) $n = 1$ 3) $n^2 = 3$ 4) $n = -1$

$$3. \sum_{k=1}^n k \left(1 + \frac{1}{n} \right)^{k-1} =$$

$$1) n(n-1) \quad 2) n(n+1) \quad 3) n^2 \quad 4) (n+1)^2$$

4. Sum of n^{th} bracket of

(1) + (2+3+4) + (5+6+7+8+9) + is

$$1) (n-1)^3 + n^3 \quad 2) (n-1)^3 + 8n^2$$

$$3) \frac{(n+1)(n+2)}{6} \quad 4) \frac{(n+3)(n+2)}{12}$$

$$5. \frac{1^2}{1} + \frac{1^2+2^2}{1+2} + \frac{1^2+2^2+3^2}{1+2+3} + \dots + n \text{ terms} =$$

$$1) \frac{n(n+3)}{4} \quad 2) \frac{n(n+3)}{5}$$

$$3) \frac{n(n+2)}{3} \quad 4) \frac{n(n+5)}{6}$$

$$6. \sum_{k=1}^5 \frac{1^3 + 2^3 + \dots + k^3}{1+3+5+\dots+(2k-1)} =$$

$$1) 22.5 \quad 2) 24.5 \quad 3) 28.5 \quad 4) 32.5$$

7. $S_n = 1^3 + 2^3 + 3^3 + \dots + n^3$ and

$$T_n = 1 + 2 + 3 + 4 + \dots + n \quad [\text{Eam-2007}]$$

$$1) S_n = T_n^3 \quad 2) S_n = T_n^2$$

$$3) S_n = T_n^2 \quad 4) S_n = T_n^3$$

8. A = sum of first 10 natural numbers,

B = sum of squares of first 10

natural numbers, C = sum of cubes of first

10 natural numbers, D = sum of first 10

even natural numbers, then increasing

order of A, B, C, D

$$1) A, B, C, D \quad 2) A, C, D, B$$

$$3) A, B, D, C \quad 4) A, D, B, C$$

$$9. \text{ If } \sum_{r=1}^n t_r = \frac{1}{12} n(n+1)(n+2) \text{ then } \sum_{r=1}^n \frac{1}{t_r}$$

$$1) \frac{2n}{n+1} \quad 2) \frac{4n}{n+1} \quad 3) \frac{3n}{n+2} \quad 4) \frac{3n}{n+1}$$

$$10. 1 + (1+3) + (1+3+5) + \dots n \text{ brackets} =$$

$$1) \frac{n(n+1)(n+2)}{6}$$

$$2) \frac{n(n+1)(3n^2 + 23n + 46)}{12}$$

$$3) \frac{n(27n^3 + 90n^2 + 45n - 50)}{4}$$

$$4) \frac{n(n+1)(2n+1)}{6}$$

11. $\frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \dots$ upto n terms =

1) $\frac{n}{n+1}$ 2) $\frac{n}{2(n+1)}$ 3) $\frac{2n}{n+1}$ 4) $\frac{2}{n(n+1)}$

Divisibility:

12. $\forall n \in \mathbb{N}, 5^{2n+2} - 24n - 25$ is divisible by

1) 576 2) 25 3) 24 4) 50

13. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is

1) 2 2) 7 3) 8 4) 0

Inequalities :

14. $\frac{(2n)!}{2^{2n}(n!)^2} \leq$

1) $\frac{1}{\sqrt{3n+1}}$ 2) $\frac{1}{\sqrt{3n+2}}$ 3) $\frac{1}{\sqrt{3n+4}}$ 4) $\frac{1}{\sqrt{3n+5}}$

15. If $n > 1$ then

1) $n! = \left(\frac{n+1}{2}\right)^n$ 2) $n! < \left(\frac{n+1}{2}\right)^n$

3) $n! > \left(\frac{n+1}{2}\right)^n$ 4) none

LEVEL-II (H.W) KEY

- | | | | |
|-------|-------|-------|-------|
| 01) 4 | 02) 2 | 03) 3 | 04) 1 |
| 05) 3 | 06) 1 | 07) 3 | 08) 4 |
| 09) 2 | 10) 4 | 11) 3 | 12) 1 |
| 13) 1 | 14) 1 | 15) 2 | |

LEVEL-II (H.W) HINTS

1. $\sum_{n=1}^{10} n^3 - \sum_{r=1}^3 r^3$

2. Put $n = 1$ and verify the options.

3. By verification.

4. Put $n = 2$ and verify the options.

5. Put $n = 2$ and verify the options.

6. $\sum_{k=1}^5 \left(\frac{k^2(k+1)^2}{4k^2} \right) = \sum_{k=1}^5 \left(\frac{(k+1)^2}{4} \right)$

7. Put $n = 2$ and verify the options.

8. $A = \sum n, B = \sum n^2, C = \sum n^3, D = \sum 2n$

9. $t_r = \sum_{k=1}^r t_k - \sum_{k=1}^{r-1} t_k = \frac{1}{12} r(r+1)(r+2) -$

$$\frac{1}{12} (r-1)r(r+1) = \frac{r(r+1)}{4}$$

10. Conceptual

11. $t_n = \frac{\sum n}{\sum n^2} = \frac{2}{n(n+1)}$

12. Put $n = 1$ and verify the options.

13. Using modulo arithmetic

$$8 \equiv -1 \pmod{9} \text{ also } 62 \equiv -1 \pmod{9}$$

$$\Rightarrow 8^{2n} - (62)^{2n+1} = [(-1)^{2n} - (-1)^{2n+1}] \pmod{9}$$

$$= (1+1) \pmod{9}$$

$$= 2 \pmod{9} \Rightarrow \text{Remainder} = 2$$

14. If $n = 1$ then

$$\frac{(2n)!}{2^n(n!)^2} = \frac{2}{4} = \frac{1}{2},$$

$$\frac{1}{\sqrt{3n+1}} = \frac{1}{\sqrt{4}} = \frac{1}{2}, \frac{1}{\sqrt{3n+2}} = \frac{1}{\sqrt{5}},$$

$$\frac{1}{\sqrt{3n+4}} = \frac{1}{\sqrt{7}}$$

$$\frac{1}{\sqrt{3n+5}} = \frac{1}{\sqrt{8}}, \therefore \frac{(2n)!}{2^{2n}(n!)^2} \leq \frac{1}{\sqrt{3n+1}}$$

15. If $n=2$ then $n!=2, \left(\frac{n+1}{2}\right)^n = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

$$\Rightarrow n! < \left(\frac{n+1}{2}\right)^n$$

LEVEL - III

1. $\frac{1.2^2 + 2.3^2 + 3.4^2 + \dots + n(n+1)^2}{1^2.2 + 2^2.3 + 3^2.4 + \dots + n^2(n+1)} =$

1) $\frac{3n+1}{3n+5}$ 2) $\frac{3n+5}{3n+1}$

3) $(3n+1)(3n+5)$ 4) $\frac{3n+5}{3n+7}$

2. If $P(n) = 1 + 2 + 3 + \dots + n$ is a perfect square N^2 and N is less than 100, then possible values of n are

1) only 1 2) 1 & 8 3) only 8 4) 1, 8, 49

3. For any $n \in N$, the value of the expression

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots n \text{ times}}}}$$
 is

1) $2 \cos\left(\frac{\pi}{2^{n+1}}\right)$ 2) $2 \sin\left(\frac{\pi}{2^{n+1}}\right)$

3) $\sqrt{2} \cos(2^{n+1}\pi)$ 4) $2 \cos(2^n\pi)$

4. $7 + 77 + 777 + \dots + (777 \dots 7 \text{ n times}) =$

1) $\frac{7}{81}(10^{n+1} - 9n - 10)$ 2) $\frac{7}{81}(10^n - 9n - 10)$

3) $\frac{7}{81}(10^{n+1} + 9n + 10)$ 4) $\frac{7}{81}(10^{n+1} + 9n - 10)$

5. $\forall n \in N, 1 + 2x + 3x^2 + \dots + n.x^{n-1} =$
($x \in R, x \neq 1$)

1) $\frac{1 - (n+1)x^n + n.x^{n+1}}{(1-x)^2}$ 2) $\frac{(n+1)x^n}{(1-x)^2}$

3) $\frac{1 - (n+1)x^n + n.x^{n+1}}{(1+x)^2}$ 4) $\frac{(n-1)x^n}{(1+x)^2}$

6. If the sum to 'n' terms of an A.P. is $\frac{4n^2 - 3n}{4}$, then the n^{th} term of the A.P. is

1) $\frac{5n-1}{4}$ 2) $\frac{8n-7}{4}$ 3) $\frac{3n^2-2}{4}$ 4) $\frac{7n-8}{4}$

7. $1 + \left(\frac{1}{3} + \frac{1}{3^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^4} + \frac{1}{3^5}\right) + \dots$ sum of the terms in the n^{th} bracket =

1) $\frac{(3^n - 1)^3}{2.4^{n-1}}$ 2) $\frac{3^n - 1}{2.3^{(n-1)(n+2)/2}}$

3) $\frac{3^n + 1}{3.7^{n-1}}$ 4) $\frac{3^{n-1}}{2^{n+1}}$

8. $(1+x)^n - nx - 1$ is divisible by (where $n \in N$)

1) $2x$ 2) x^2 3) $2x^3$ 4) all of these

9. For a positive integer ($n > 1$),

Let $a(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{(2^n) - 1}$. Then

1) $a(100) < 100$ 2) $a(100) > 100$

3) $a(20) = 100$ 4) $a(200) \geq 100$

10. The sum of the first 'n' terms of the series

$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$ is

$\frac{n(n+1)^2}{2}$ when n is even. When n is odd

the sum is

1) $\frac{3n(n+1)}{2}$ 2) $\frac{n^2(n+1)}{2}$

3) $\frac{n(n+1)^2}{4}$ 4) $\left[\frac{n(n+1)}{2}\right]^2$

11. If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}, \forall n \in N$,

then $H_1 + H_2 + H_3 + \dots + H_n =$

1) $(n+1)H_n - n$ 2) $(n+1)H_n + n$

3) $(n+1)H_n$ 4) $(n-1)H_n - n$

12. $S_n = \frac{1+2+3+\dots+n}{n}$ then

$S_1^2 + S_2^2 + S_3^2 + \dots + S_n^2 =$

1) $\frac{n}{24}(2n^2 + 9n + 13)$ 2) $\frac{1}{24}(2n^2 + 9n + 13)$

3) $\frac{n^2}{24}(2n^2 + 9n + 13)$ 4) $\frac{n}{24}(2n^2 - 9n + 13)$

13. If $t_n = \frac{1}{4}(n+2)(n+3)$ for $n = 1, 2, 3, \dots$

then $\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \dots + \frac{1}{t_{2003}} =$

1) $\frac{4006}{3006}$ 2) $\frac{4003}{3007}$ 3) $\frac{4006}{3008}$ 4) $\frac{4006}{3009}$

14. The value of the sum in the 50th bracket of $(1) + (2+3) + (4+5+6) + (7+8+9+10) + \dots$ is

1) 62525 2) 65225 3) 56255 4) 55625

15. $1 + \frac{x}{a_1} + \frac{x(x+a_1)}{a_1 a_2} + \dots$

$+ \frac{x(x+a_1)(x+a_2)\dots(x+a_{n-1})}{a_1 a_2 \dots a_n} =$

1) $\frac{(x+a_1)(x+a_2)\dots(x+a_n)}{a_1 a_2 \dots a_n}$

2) $\frac{(x-a_1)(x-a_2)\dots(x-a_n)}{a_1 a_2 \dots a_n}$

3) $(x+a_1)(x+a_2)\dots(x+a_n)$

4) $(x-a_1)(x-a_2)\dots(x-a_n)$

16. For all positive integers $n > 1$,

$\{x(x^{n-1} - n.a^{n-1}) + a^n(n-1)\}$ is divisible by

1) $(x-a)^2$ 2) $x-a$ 3) $2(x-a)$ 4) $x+a$

17. $\forall n \in \mathbf{N}, x \in \mathbf{R}$,

$\tan^{-1}\left[\frac{x}{1.2+x^2}\right] + \tan^{-1}\left[\frac{x}{2.3+x^2}\right] + \dots +$

$\tan^{-1}\left[\frac{x}{n(n+1)+x^2}\right] =$

1) $\tan^{-1}\left[\frac{x}{n}\right] - \tan^{-1}\left[\frac{x}{n+1}\right]$

2) $\tan^{-1}[x] - \tan^{-1}\left[\frac{x}{n+1}\right]$

3) $\tan^{-1}[n+1] - \tan^{-1}[x]$ 4) $\tan^{-1}[x]$

18. $\tan^{-1}\left(\frac{1}{1+1+1^2}\right) + \tan^{-1}\left(\frac{1}{1+2+2^2}\right)$

$+ \dots + \tan^{-1}\left(\frac{1}{1+n+n^2}\right) =$

1) $\tan^{-1}(n+1) + \pi$ 2) $\tan^{-1}(n+1) + \frac{\pi}{4}$

3) $\tan^{-1}(n+1)$ 4) $\tan^{-1}(n+1) - \frac{\pi}{4}$

19. $\frac{1}{2}\tan\left(\frac{x}{2}\right) + \frac{1}{4}\left(\tan\frac{x}{4}\right) + \dots + \frac{1}{2^n}\tan\left(\frac{x}{2^n}\right) =$

1) $\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right)$ 2) $\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right) + \cot x$

3) $\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right) - \cot x$ 4) $\cot\left(\frac{x}{2^n}\right) - \cot x$

20. Sum of first 'n' terms of the series

$\frac{3}{2} + \frac{5}{4} + \frac{9}{8} + \frac{17}{16} + \dots$

1) $n-1+2^{-n}$ 2) $n+1+2^{-n}$

3) $n+1-2^{-n}$ 4) $n-1-2^{-n}$

21. If n is even, then the sum of first 'n' terms of the series

$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$ is

1) $\frac{n(n+1)^2}{3}$ 2) $\frac{n(n+1)^2}{4}$

3) $\frac{n(n+1)^2}{2}$ 4) $\frac{n^2(n+1)}{2}$

22. If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$, then the value of

$$1 + \frac{3}{2} + \frac{5}{3} + \dots + \frac{2n-1}{n} \text{ is}$$

- 1) $H_n + n$ 2) $2n - H_n$
3) $n - 1 + H_n$ 4) $H_n + 2n$

23. If $S_1 = \{2\}$, $S_2 = \{3, 6\}$, $S_3 = \{4, 8, 16\}$,
 $S_4 = \{5, 10, 20, 40\}$, then the sum of
numbers in the set S_{15} is

- 1) $5(2^{15})$ 2) $16(2^{15} - 1)$
3) $16(2^{16} - 1)$ 4) $15(2^{15} - 1)$

24. The sum of first 'n' terms of the series
 $1^2 + (1)(2) + 3^2 + (3)(4) + 5^2 + (5)(6) + 7^2 + \dots$
when n is odd is

- 1) $\frac{1}{12}(n+1)(4n^2 - n + 3)$
2) $\frac{1}{12}n(4n^2 + 3n - 4)$
3) $\frac{1}{6}(n+1)(4n^2 - n + 5)$
4) $\frac{1}{6}(n)(4n^2 + 5n - 6)$

25. The sets $S_1, S_2, S_3, \dots$ are given by

$$S_1 = \left\{ \frac{2}{1} \right\}, S_2 = \left\{ \frac{3}{2}, \frac{5}{2} \right\},$$

$$S_3 = \left\{ \frac{4}{3}, \frac{7}{3}, \frac{10}{3} \right\}, S_4 = \left\{ \frac{5}{4}, \frac{9}{4}, \frac{13}{4}, \frac{17}{4} \right\}, \dots$$

then the sum of the numbers in the set S_{25} is

- 1) 322 2) 324 3) 325 4) 326

26. If $S_1 = \{2\}$, $S_2 = \left\{ \frac{3}{2}, \frac{4}{2} \right\}$, $S_3 = \left\{ \frac{4}{4}, \frac{5}{4}, \frac{6}{4} \right\}$,

$$S_4 = \left\{ \frac{5}{8}, \frac{6}{8}, \frac{7}{8}, \frac{8}{8} \right\}, \dots \text{ then the sum of}$$

numbers in S_{20} is

- 1) $\frac{1220}{2^{20}}$ 2) $\frac{1563}{2^{19}-1}$ 3) $\frac{1445}{2^{18}}$ 4) $\frac{1576}{2^{21}}$

27. Sum of the series

$$S = 1^2 - 2^2 + 3^2 - 4^2 + \dots - 2002^2 + 2003^2 \text{ is}$$

- 1) 2007006 2) 1005004
3) 2000506 4) none

28. The positive integer 'n' for which

$$2 \times 2^2 + 3 \times 2^3 + 4 \times 2^4 + \dots + n \times 2^n = 2^{n+10} \text{ is}$$

- 1) 510 2) 511 3) 512 4) 513

29. $\sum_{k=1}^{2n+1} (-1)^{k-1} k^2 =$

- 1) $(n+1)(2n+1)$ 2) $(n+1)(2n-1)$
3) $(n-1)(2n-1)$ 4) $(n-1)(2n+1)$

Divisibility:

30. Let $P(n) = 5^n - 2^n$, $P(n)$ is divisible by 3λ
where λ and n both are odd positive integers then the least value of 'n' and λ will be

- 1) 13 2) 11 3) 1 4) 5

31. The least remainder when 17^{30} is divided by 5 is

- 1) 1 2) 2 3) 3 4) 4

32. When 2^{301} is divided by 5, the least +ve remainder is

- 1) 4 2) 8 3) 2 4) 6

33. The remainder when 5^{99} is divided by 13 is

- 1) 6 2) 8 3) 9 4) 10

LEVEL - III-KEY

01) 2	02) 4	03) 1	04) 1
05) 1	06) 2	07) 2	08) 2
09) 1	10) 2	11) 1	12) 1
13) 4	14) 1	15) 1	16) 1
17) 2	18) 4	19) 3	20) 3
21) 3	22) 2	23) 2	24) 1
25) 4	26) 1	27) 1	28) 4
29) 1	30) 3	31) 4	32) 3
33) 2			

LEVEL - III-HINTS

- Put $n = 2$ and verify the options.
- By Observation

3. By Verification
4. Put $n = 1$ and verify the options.
5. Put $n = 2$ and verify the options.
6. $t_n = S_n - S_{n-1}$
7. By Verification
8. Put $n = 2$, and $x = 3$
not divisible by 6, 54 but divisible by 9.

9. $\frac{n}{2} < a(n) < n$

10. By verification

11. By Verification

12. Put $n = 2$ and verify the options.

$$S_1^2 + S_2^2 = 1 + \frac{9}{4} = \frac{13}{4}$$

13. $\frac{1}{t_n} = \frac{4}{(n+2)(n+3)} = 4 \left[\frac{1}{(n+2)} - \frac{1}{(n+3)} \right]$

14. First term of 50th bracket
 $= (1+2+3+ \dots + 49) + 1 = 1226$

15. Put $n = 1$ and verify the options.

16. Put $n = 2$ and verify the options.

17. Put $n = 1$ and verify the options.

18. $\tan^{-1} \left(\frac{1}{1+n+n^2} \right) = \tan^{-1}(n+1) - \tan^{-1}(n)$

$$\tan^{-1} \left(\frac{1}{1+1+1^2} \right) + \tan^{-1} \left(\frac{1}{1+2+2^2} \right)$$

$$+ \dots + \tan^{-1} \left(\frac{1}{1+n+n^2} \right)$$

$$= \tan^{-1}(n+1) - \tan^{-1}(1) = \tan^{-1}(n+1) - \frac{\pi}{4}$$

19. Put $n = 1$ and verify the options.

20. Put $n = 2$ and verify the options.

21. By verification

22. $(2-1) + \left(2 - \frac{1}{2}\right) + \left(2 - \frac{1}{3}\right) + \dots$

$$= 2n - \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots\right) = 2n - H_n$$

23. $T_n = (n+1)(2^n - 1)$

24. Put $n = 3$ and verify

25. $S_{25} = \left\{ \frac{26}{25}, \frac{51}{25} \dots 25 \text{ terms} \right\}$

26. Ist term in $S_1 = \frac{2}{2^0}$, Ist term in $S_2 = \frac{3}{2^1}$,

Ist term in $S_3 = \frac{4}{2^2}$, Ist term in $S_{20} = \frac{21}{2^{19}}$

$$\text{Sum in } S_{20} = \frac{20}{2} \left[2 \times \frac{21}{2^{19}} + (20-1) \frac{1}{2^{19}} \right]$$

27. We can write S as

$$S = (1-2)(1+2) + (3-4)(3+4) + \dots + (2001-2002)(2001+2002) + 2003^2$$

$$= -[1+2+3+4+\dots+2002] +$$

$$2003^2 = -\frac{1}{2}(2002)(2003) + 2003^2 = 2007006$$

28. We have $2^{n+10} = 2.2^2 + 3.2^3 + 4.2^4 + \dots +$

$$n.2^n \Rightarrow 2(2^{n+10}) = 2.2^3 + 3.2^4 + \dots +$$

$$+(n-1).2^n + n.2^{n+1}$$

subtracting, we get

$$-2^{n+10} = 2.2^2 + 2^3 + 2^4 + \dots + 2^n - n.2^{n+1} = 8$$

$$+ \frac{8(2^{n-2} - 1)}{2-1} - n.2^{n+1}$$

$$= 8 + 2^{n+1} - 8 - n.2^{n+1} = 2^{n+1} - (n)2^{n+1}$$

$$\Rightarrow 2^{10} = 2n - 2 \Rightarrow n = 513$$

29. $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + (2n+1)^2$

$$= 1^2 + 2^2 + 3^2 + \dots + (2n+1)^2$$

$$- 2 \left[2^2 + 4^2 + \dots + (2n)^2 \right]$$

$$= \frac{(2n+1)(2n+2)(4n+3)}{6}$$

$$- \frac{8.n(n+1)(2n+1)}{6} = (n+1)(2n+1)$$

30. $P(n) = 5^n - 2^n$

$$n = 1, \therefore P(1) = 3\lambda = 3 \therefore \lambda = 1$$

$$\text{Similarly } n = 5, \therefore P(5) = 5^5 - 2^5 \\ = 3125 - 32 = 3093 = 3 \times 1031$$

In this case $\lambda = 1031$

Similarly we can check the result for other cases and find that the least value of λ and n is 1.

$$31. \quad 17 \equiv 2 \pmod{5}$$

$$17^5 \equiv 2^5 \pmod{5} \equiv 2 \pmod{5} \\ \Rightarrow (17^5)^6 = 2^6 \pmod{5} \Rightarrow (17)^{30} \equiv 4 \pmod{5}$$

$$32. \quad 2^4 \equiv 1 \pmod{5} \Rightarrow (2^4)^{75} \equiv (1)^{75} \pmod{5} \text{ i.e.,} \\ 2^{300} \equiv 1 \pmod{5} \Rightarrow 2^{300} \times 2 \equiv (1 \cdot 2) \pmod{5} \\ \Rightarrow 2^{301} \equiv 2 \pmod{5} \therefore \text{Least +ve remainder is 2.}$$

$$33. \quad 5^{99} = (5)(5^2)^{49} = 5(25)^{49} = 5(26-1)^{49} \\ = 5 \times 26 \times (\text{positive term}) - 5, \\ \text{so when it is divided by 13 its gives the remain-} \\ \text{der ' -5 ' or } (13-5) \text{ i.e., 8.}$$



LEVEL - IV

1. **Statement -1:** For all $n \in N$, $x^{2n+1} + y^{2n+1}$ is divisible by $x + y$

Statement - 2: If $n \in N$, $n^3 + 2n$ is divisible by 6

Which of the above statement is true:

- 1) only 1 2) only 2
3) both 1 & 2 4) neither 1 nor 2

2. $49^n + 16n - 1$ is divisible by $A(n \in N)$

$3^{2n} + 7$ is divisible by $B(n \in N)$

$4^n - 3n - 1$ is divisible by $C(n \in N)$

$3^{3n} - 26n - 1$ is divisible by $D(n \in N)$

then the increasing order of A, B, C, D is

- 1) A, B, C, D 2) C, B, A, D
3) B, C, A, D 4) D, A, C, B

3. **Statement - 1:** For all $n \in N$ $x^n - y^n$ is divisible by $x - y$

Statement - 2: $x^n + y^n$ is divisible by $x + y$ if n is even natural number

Which of the above statement is true :

- 1) only 1 2) only 2
3) both 1 & 2 4) neither 1 nor 2

4. **Assertion (A):** For all +ve, integral values of 'n', $3^{2n} + 7$ is divisible by 8

Reason (R) : G.C.F. of 16 and 88 is 8

- 1) A true, R true and R is the correct explanation of A
2) A true, R true and R is not a correct explanation of A
3) A true, R false 4) A false, R true.

5. **Assertion (A):** $n \in N$ product

of $n(n+1)(n+2)$ is divisible by 6

Reason (R): Product of 3 consecutive +ve integers is divisible by 3!

- 1) A true, R true and R is the correct explanation of A
2) A true, R true and R is not a correct explanation of A
3) A true, R false 4) A false, R true.

6. **Statement -1:** For every natural number

$$n \geq 2, \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}.$$

Statement - 2 :

For every natural number $n \geq 2$,

$$\sqrt{n(n+1)} < n+1. \quad [\text{AIE- 2008}]$$

- 1) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement-1
2) Statement - 1 is true, Statement - 2 is false
3) Statement - 1 is false, Statement - 2 is true
4) Statement - 1 is true, Statement - 2 is true ;

Statement - 2 is a correct explanation for Statement-1

7. **Assertion (A):**

$$n(n^2 - 1^2)(n^2 - 2^2) \cdots (n^2 - r^2), n > r, n \in N$$

is divisible by $(2r+1)!$

Reason (R) : Product of 'r' consecutive natural numbers is divisible by $r!$.

1) Both A and R are true and R is the correct Explanation of A

2) Both A and R are true but R is not the correct Explanation of A

3) A is true and R is false

4) A is false and R is true

8. **Statement-1:** The digit in the unit place of $183! + 3^{183}$ is 7.

Statement-2: $183!$ have unit place 0 and 3^{4k+3} ends with 7.

1) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1.

2) Statement-1 is true, Statement-2 is false.

3) Statement-1 is false, Statement-2 is true.

4) Statement-1 is true, Statement-2 is true, Statement-2 is correct explanation for Statement-1.

9. **Statement-1:** If $A = (300)^{600}$, $B = 600!$,

$$C = (200)^{600}, \text{ then } A > B > C.$$

Statement-2: $\left(\frac{n}{2}\right)^n > n! > \left(\frac{n}{3}\right)^n$ for $n > 6$.

1) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1

2) Statement-1 is true, Statement-2 is false.

3) Statement-1 is false, Statement-2 is true.

4) Statement-1 is true, Statement-2 is true,

Statement-2 is correct explanation for Statement-1.

10. **Statement-1:** Let

$$P(x) = (x+1)(x+2)(x+3) \cdots (x+11),$$

then coefficient of $x^{10} = 64$.

Statement-2: Coefficient of x^{n-1} in the product of

$$(x+1)(x+2) \cdots (x+n) \text{ is } \frac{n(n+1)}{2}.$$

1) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1

2) Statement-1 is true, Statement-2 is false.

3) Statement-1 is false, Statement-2 is true.

4) Statement-1 is true, Statement-2 is true, Statement-2 is correct explanation for Statement-1.

11. **Statement - 1 :-** The sum of the series

$$1 + (1+2+4) + (4+6+9) + (9+12+16) + \cdots + (361+380+400) \text{ is } 8000$$

$$\text{Statement - 2 :- } \sum_{k=1}^n [k^3 - (k-1)^3] = n^3,$$

for any natural number 'n'

1) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1

2) Statement-1 is true, Statement-2 is false

3) Statement-1 is false, Statement-2 is true

4) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1

12. **Statement - 1 :** For each natural number n,

$$(n+1)^7 - n^7 - 1 \text{ is divisible by } 7.$$

Statement - 2 : For each natural number n,

$n^7 - n$ is divisible by 7. (AIEEE-2011)

- 1) Statement -1 is false, Statement-2 is true
- 2) Statement-1 is true, Statement -2 is true & Statement -2 is correct explanation of Statement-1
- 3) Statement-1 is true, Statement -2 is true & Statement -2 is not a correct explanation of Statement-1
- 4) Statement-1 is true, Statement-2 is false.

LEVEL - IV-KEY

01) 1	02) 3	03) 1	04) 1
05) 1	06) 1	07) 1	08) 4
09) 4	10) 3	11) 4	12) 2

LEVEL - IV-HINTS

1. Put $n = 1$ and verify the options.
2. Put $n = 1, 2, 2, 2$, $B = 4$, $C = 9$, $A = 64$, $D = 666$
3. Put $n = 1, 2$ and verify the options.
4. Put $n = 1, 2$ and take G.C.F
5. Product of 3 successive integers is divisible by $3!$.
6. By verification
7. By observation
8. For every $n!$ $\forall n \geq 5$ ends with 0 and 3^{4n} ends with 1.
 $\therefore 3^{4n+3} = 3^{4n} \cdot 3^3$ where unit place is 7.
9. Put $n = 600 \Rightarrow \left(\frac{600}{2}\right)^{600} > 600! > (200)^{600}$
 $\Rightarrow A > B > C$
10. $(x+1)(x+2)\dots(x+n)$
 $x^n + (1+2+\dots+n)x^{n-1} +$
 $[(1 \cdot 2 + 1 \cdot 3 + \dots + 1 \cdot n) +$
 $(2 \times 3 + \dots + 2 \times n) + \dots + (n+1) \times n]x^{n-2}$

$$+ \dots + (1 \times 2 \times \dots \times n) = \frac{n(n+1)}{2} \Rightarrow \text{coefficient of}$$

$$x^{10} = x^{11-1} = \frac{11(12)}{2} = 66 \neq 64$$

$$11. \sum_{k=1}^n [k^3 - (k-1)^3] =$$

$$(1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \dots +$$

$$(n^3 - (n-1)^3) = n^3$$

$$1 + (1+2+4) + (4+6+9) + (9+12+16) + \dots +$$

$$(361 + 380 + 400) =$$

$$(1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \dots +$$

$$(20^3 - 19^3) = 20^3 = 8000$$

$$12. \text{ For } n = k, P(k) = k^7 - k$$

Let $P(k)$ is divisible by 7.

$$k^7 - k = 7\lambda \text{ for some } \lambda \in N \dots (1) \text{ For}$$

$$n = K+1, P(k+1) = (k+1)^7 - (K+1)$$

$$= ({}^7C_0 k^7 + {}^7C_1 k^6 + {}^7C_2 k^5 + \dots + {}^7C_6 k + {}^7C_7)$$

$$- (K+1) = (K^7 - k) + 7(k^6 + 3k^5 + \dots + k)$$

$$= 7\lambda + 7(k^6 + 3k^5 + \dots + k)$$